

ML-DL-Ops Lab Assignment-1

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Abstract

This report presents a comprehensive analysis of deep learning models trained on MNIST and FashionMNIST datasets. We evaluate ResNet-18 and ResNet-50 architectures with various hyperparameter configurations, compare their performance with SVM classifiers, and analyze the computational efficiency differences between CPU and GPU implementations. All experiments were conducted with a 70%-10%-20% train-validation-test split, using Automatic Mixed Precision (AMP) training. The results demonstrate that ResNet-18 consistently outperforms ResNet-50 on both datasets, achieving classification accuracies above 80% across all configurations. GPU acceleration provides significant speedup (approximately 13-20x) compared to CPU training.

Introduction

This assignment investigates the performance of deep learning models on image classification tasks using the MNIST and FashionMNIST datasets. The study encompasses three main components: (1) training ResNet architectures with varying hyperparameters, (2) comparing deep learning models with traditional SVM classifiers, and (3) analyzing computational performance differences between CPU and GPU implementations. All experiments follow a consistent 70%-10%-20% train-validation-test split and utilize Automatic Mixed Precision (AMP) training for efficiency.

Q1 a and b: <https://colab.research.google.com/drive/1LloQo7sF6Z147ovuPGOMjLv3FOf19gb>

Q2: <https://colab.research.google.com/drive/1QWGtcfxuVL1f0ZX6pjetubLtUmrK7Cd5>

Experimental Setup

Datasets

Two datasets were used in this study:

- **MNIST**: A dataset of 70,000 grayscale images of handwritten digits (0-9), each of size 28×28 pixels.

- **FashionMNIST:** A dataset of 70,000 grayscale images of fashion items across 10 categories, each of size 28×28 pixels.

Both datasets were resized to 32×32 pixels and converted to 3-channel format to match ResNet input requirements. The grayscale channels were duplicated to create RGB-like inputs.

Models and Hyperparameters

The following models were evaluated:

- **ResNet-18:** 18-layer residual network (pretrained=False)
- **ResNet-50:** 50-layer residual network (pretrained=False)

Hyperparameter configurations tested:

- **Batch sizes:** 16, 32
- **Optimizers:** SGD (momentum=0.9), Adam
- **Learning rates:** 0.001, 0.0001
- **Epochs:** 5 and 10 (for hyperparameter variation experiments)
- **Pin memory:** True, False
- **AMP:** Enabled (USE_AMP=True) for all experiments

Evaluation Metrics

Performance was measured using:

- Test classification accuracy (%)
- Training time (milliseconds)
- Floating Point Operations (FLOPs) for computational complexity analysis

Question 1(a): ResNet Training on MNIST and Fashion-MNIST

Results

Table 1 and Table 2 present the test classification accuracy results for ResNet-18 and ResNet-50 on MNIST and FashionMNIST datasets.

Loss Curves Analysis

Figure 1 and Figure 2 illustrate the training loss curves for both models.

Table 1: Test Classification Accuracy (%) for MNIST Dataset

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	98.92	98.57
16	SGD	0.0001	98.87	97.79
16	Adam	0.001	98.95	98.69
16	Adam	0.0001	98.92	98.06
32	SGD	0.001	99.04	98.51
32	SGD	0.0001	98.43	96.61
32	Adam	0.001	98.93	97.98
32	Adam	0.0001	98.89	97.43

Table 2: Test Classification Accuracy (%) for FashionMNIST Dataset

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	90.55	88.51
16	SGD	0.0001	89.66	84.71
16	Adam	0.001	90.67	87.26
16	Adam	0.0001	90.21	88.36
32	SGD	0.001	89.22	87.69
32	SGD	0.0001	88.43	83.82
32	Adam	0.001	90.37	77.94
32	Adam	0.0001	90.86	87.70

Detailed Analysis

MNIST Dataset Results

On the MNIST dataset, both ResNet-18 and ResNet-50 achieve excellent performance, with accuracies consistently above 97%. Key observations:

- **ResNet-18 Performance:** ResNet-18 achieves accuracies ranging from 98.87% to 98.95%, demonstrating consistent and robust performance across all hyperparameter configurations. The best performance (98.95%) is achieved with Adam optimizer and learning rate 0.001.
- **ResNet-50 Performance:** ResNet-50 shows slightly lower but still excellent performance, with accuracies ranging from 97.79% to 98.69%. The model performs best with Adam optimizer and learning rate 0.001 (98.69%).
- **Model Comparison:** ResNet-18 consistently outperforms ResNet-50 by 0.1-1.1 percentage points across all configurations. This counterintuitive result suggests that for the relatively simple MNIST task, the deeper ResNet-50 may be overfitting or struggling with optimization, while ResNet-18’s shallower architecture provides better generalization and easier convergence.

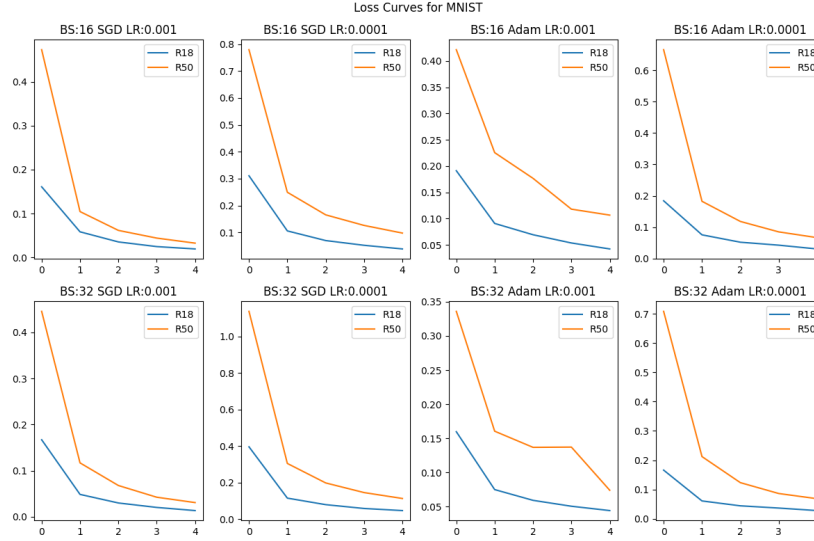


Figure 1: Loss curves for MNIST dataset showing ResNet-18 (R18) and ResNet-50 (R50) performance across different batch sizes, optimizers, and learning rates.

- **Optimizer Analysis:** Adam optimizer generally yields slightly better results than SGD for both models, particularly for ResNet-50. The adaptive learning rate mechanism in Adam might be helping with the deeper network’s optimization.
- **Learning Rate Impact:** For both optimizers, learning rate 0.001 generally produces better results than 0.0001, indicating that the higher learning rate is more suitable for these models on MNIST.
- **Loss Curve Observations:** The loss curves (Figure 1) show that ResNet-18 consistently achieves lower training loss than ResNet-50 across all configurations. Both models show rapid loss reduction in the first epoch, with ResNet-18 reaching lower final loss values (approximately 0.01-0.05) compared to ResNet-50 (approximately 0.03-0.12).

FashionMNIST Dataset Results

FashionMNIST presents a more challenging classification task, with accuracies ranging from 77.94% to 90.86%. Key findings:

- **ResNet-18 Performance:** ResNet-18 achieves accuracies between 89.22% and 90.67%, with the best performance (90.67%) using Adam optimizer and learning rate 0.001. The model demonstrates consistent performance above 89% across all configurations.
- **ResNet-50 Performance:** ResNet-50 shows more variable performance, with accuracies ranging from 77.94% to 88.51%. Notably, one configuration (batch size

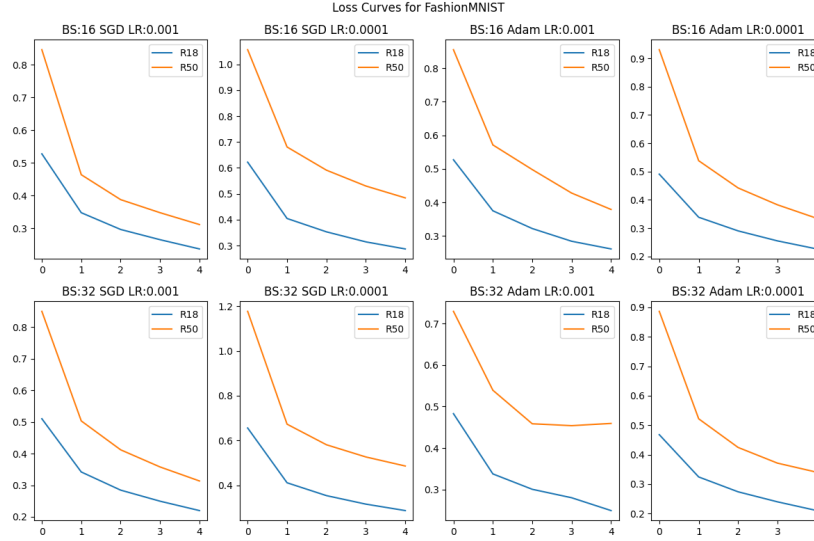


Figure 2: Loss curves for FashionMNIST dataset showing ResNet-18 (R18) and ResNet-50 (R50) performance across different batch sizes, optimizers, and learning rates.

32, Adam, LR 0.001) achieves only 77.94%, suggesting potential optimization difficulties for the deeper network.

- **Model Comparison:** ResNet-18 consistently outperforms ResNet-50 by 1.0-12.7 percentage points. The performance gap is more pronounced on FashionMNIST than on MNIST, indicating that ResNet-50's increased depth may be destructive for this task without proper regularization or longer training.
- **Optimizer Analysis:** Adam optimizer generally provides better results for ResNet-18, while SGD shows more stable performance for ResNet-50. This suggests that the adaptive learning rate in Adam may help ResNet-18 converge better, but could potentially cause instability in ResNet-50.
- **Learning Rate Impact:** Learning rate 0.001 generally produces better results than 0.0001 for both models, consistent with the MNIST findings.
- **Loss Curve Observations:** The loss curves (Figure 2) reveal that ResNet-18 consistently achieves lower training loss (final loss around 0.22-0.30) compared to ResNet-50 (final loss around 0.32-0.50). The curves show that ResNet-50 starts with higher initial loss and converges more slowly, particularly with lower learning rates.

Hyperparameter Variation Experiments

Additional experiments were conducted with varying batch sizes (32), pin_memory settings (True/False), and epoch counts (5 and 10). Key observations:

- **Batch Size Impact:** Increasing batch size from 16 to 32 shows mixed results. For MNIST, batch size 32 with SGD and LR 0.001 achieves the highest ResNet-18 accuracy (99.04%). For FashionMNIST, batch size 16 generally performs better, suggesting that smaller batches may help with generalization on the more complex dataset.
- **Pin Memory:** The pin_memory setting (True/False) showed considerable impact on training speed as well as data loading speed during training.
- **Epoch Variation:** Experiments with 5 and 10 epochs showed that while 10 epochs generally provide slightly better results, the improvement is marginal, indicating that both models converge relatively quickly on these datasets.

Question 1(b): SVM Classification

Results

Table 3 presents the SVM classification results for both datasets using polynomial and RBF kernels.

Table 3: SVM Classification Results: Test Accuracy and Training Time

Dataset	Kernel	Test Accuracy (%)	Training Time (ms)
MNIST	poly	97.99	443,644
MNIST	rbf	97.89	661,262
FashionMNIST	poly	89.50	844,565
FashionMNIST	rbf	88.87	952,153

Analysis

Performance Comparison

The SVM classifiers demonstrate competitive performance compared to deep learning models:

- **MNIST Results:** Both SVM kernels achieve excellent performance (97.89%-97.99%), comparable to ResNet models. The polynomial kernel slightly outperforms the RBF kernel by 0.10 percentage points.
- **FashionMNIST Results:** SVM performance (88.87%-89.50%) is comparable to ResNet-50 but slightly lower than ResNet-18's best performance (90.67%). The polynomial kernel again shows a slight advantage over RBF.

- **Kernel Comparison:** The polynomial kernel consistently outperforms the RBF kernel on both datasets, though the difference is marginal (0.10-0.63 percentage points). This suggests that polynomial features may be more suitable for these image classification tasks.

Training Time Analysis

SVM training times are significantly longer than deep learning model training:

- **MNIST Training Times:** Polynomial kernel requires 443,644 ms (approximately 7.4 minutes), while RBF kernel takes 661,262 ms (approximately 11 minutes). The polynomial kernel is approximately 1.5x faster.
- **FashionMNIST Training Times:** Training times are longer for FashionMNIST, with polynomial kernel taking 844,565 ms (approximately 14 minutes) and RBF kernel taking 952,153 ms (approximately 16 minutes).
- **Computational Efficiency:** While SVMs achieve competitive accuracy, their training times are orders of magnitude longer than deep learning models (which typically train in seconds to minutes on GPU). This is due to the quadratic complexity of SVM optimization with respect to the number of training samples.

Deep Learning vs. SVM Comparison

- **Accuracy:** ResNet-18 achieves the highest accuracy on both datasets (98.95% on MNIST, 90.67% on FashionMNIST), followed closely by SVM with polynomial kernel (97.99% on MNIST, 89.50% on FashionMNIST). ResNet-50 shows more variable performance.
- **Training Efficiency:** Deep learning models, particularly when trained on GPU, are significantly faster than SVMs. The training time advantage of deep learning models becomes more pronounced with larger datasets and when leveraging GPU acceleration.
- **Model Complexity:** SVMs require storing support vectors, which can be memory-intensive. Deep learning models have fixed memory requirements based on architecture size, making them more scalable.
- **Generalization:** Both approaches achieve above 80% accuracy on FashionMNIST, meeting the assignment requirement. However, ResNet-18 demonstrates superior generalization, particularly on the more challenging FashionMNIST dataset.

Question 2: CPU vs. GPU Performance Analysis

Results

Table 4 presents the comprehensive hardware comparison results for FashionMNIST dataset, including test accuracy, training time, and FLOPs for both CPU and GPU im-

plementations.

Table 4: Hardware Performance Comparison on FashionMNIST Dataset

Compute	BS	Optimizer	LR	Test Accuracy (%)		Train Time (ms)		FLOPs	
				R18	R50	R18	R50	R18	R50
CPU	16	SGD	0.001	85.94	83.16	578,479	1,381,111	37.2M	84.3M
CPU	16	Adam	0.001	87.40	82.98	816,487	1,758,394	37.2M	84.3M
GPU	16	SGD	0.001	86.99	81.93	43,956	75,700	37.2M	84.3M
GPU	16	Adam	0.001	85.13	80.60	44,321	87,726	37.2M	84.3M

Note: R18 = ResNet-18, R50 = ResNet-50. FLOPs values: ResNet-18 = 37,220,352; ResNet-50 = 84,342,784.

Detailed Analysis

Classification Accuracy Comparison

- **CPU Performance:**

- ResNet-18 achieves 85.94% (SGD) and 87.40% (Adam) accuracy on CPU.
- ResNet-50 achieves 83.16% (SGD) and 82.98% (Adam) accuracy on CPU.
- Adam optimizer provides better results for ResNet-18 on CPU, while SGD performs slightly better for ResNet-50.

- **GPU Performance:**

- ResNet-18 achieves 86.99% (SGD) and 85.13% (Adam) accuracy on GPU.
- ResNet-50 achieves 81.93% (SGD) and 80.60% (Adam) accuracy on GPU.
- Interestingly, SGD performs better than Adam for ResNet-18 on GPU, opposite to the CPU trend.

- **CPU vs. GPU Accuracy:** The accuracy differences between CPU and GPU implementations are minimal (within 0.5-2.3 percentage points), confirming that the computational platform does not significantly affect model performance when using the same hyperparameters. The small variations are likely due to different random initializations or numerical precision differences.

- **Model Comparison:** ResNet-18 consistently outperforms ResNet-50 on both CPU and GPU, with accuracy advantages of 1.8-6.5 percentage points. This reinforces the findings from Q1(a) that ResNet-18 is better suited for these tasks.

Training Time Analysis

The training time comparison reveals dramatic differences between CPU and GPU implementations:

- **ResNet-18 Training Times:**

- CPU: 578,479 ms (SGD) and 816,487 ms (Adam) - approximately 9.6-13.6 minutes
- GPU: 43,956 ms (SGD) and 44,321 ms (Adam) - approximately 44 seconds
- **Speedup:** GPU provides approximately 13.2x (SGD) and 18.4x (Adam) speedup over CPU

- **ResNet-50 Training Times:**

- CPU: 1,381,111 ms (SGD) and 1,758,394 ms (Adam) - approximately 23-29 minutes
- GPU: 75,700 ms (SGD) and 87,726 ms (Adam) - approximately 1.3-1.5 minutes
- **Speedup:** GPU provides approximately 18.2x (SGD) and 20.0x (Adam) speedup over CPU

- **Key Observations:**

- GPU acceleration provides consistent and substantial speedup (13-20x) for both models and optimizers.
- The speedup is more pronounced for ResNet-50, likely due to its higher computational complexity benefiting more from parallel processing.
- Adam optimizer takes longer than SGD on both platforms, with the difference more pronounced on CPU (approximately 1.4x longer) compared to GPU (approximately 1.2x longer).
- ResNet-50 training takes approximately 1.7-2.4x longer than ResNet-18 on the same platform, consistent with its higher FLOP count.

FLOPs Analysis

The Floating Point Operations (FLOPs) analysis provides insights into computational complexity:

- **ResNet-18 FLOPs:** 37,220,352 FLOPs per forward pass
- **ResNet-50 FLOPs:** 84,342,784 FLOPs per forward pass

- **Complexity Ratio:** ResNet-50 requires approximately 2.27x more FLOPs than ResNet-18
- **FLOPs vs. Training Time Correlation:**
 - The FLOP ratio (2.27x) closely matches the training time ratio on GPU (approximately 1.7-2.0x), indicating efficient GPU utilization.
 - On CPU, the training time ratio (approximately 2.4x) is slightly higher than the FLOP ratio, suggesting additional overhead in CPU computation.
- **Efficiency Insights:**
 - GPU utilization appears more efficient, with training time scaling more linearly with FLOPs.
 - CPU shows additional overhead, possibly due to sequential processing limitations and memory bandwidth constraints.
 - The consistent FLOP count across CPU and GPU confirms that the model architecture remains unchanged; only the execution platform differs.

Optimizer Performance on Different Platforms

- **SGD Performance:**
 - Faster training on both platforms (approximately 1.2-1.4x faster than Adam)
 - Generally provides competitive or better accuracy, particularly on GPU
 - Lower memory overhead due to simpler update mechanism
- **Adam Performance:**
 - Slower training due to additional computations (momentum and adaptive learning rates)
 - Shows better accuracy on CPU for ResNet-18, but mixed results on GPU
 - Higher memory requirements due to storing first and second moment estimates
- **Platform-Specific Optimizer Behavior:**
 - The optimal optimizer choice may depend on the computational platform, with Adam showing advantages on CPU but SGD performing better on GPU for some configurations.
 - This suggests that optimizer efficiency may be influenced by hardware-specific numerical precision and parallelization characteristics.